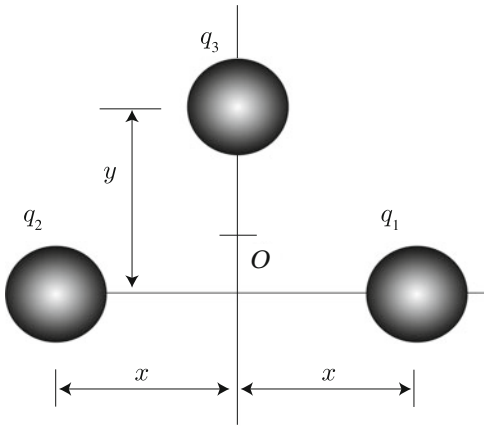


Fig. 1.21 Coordinates in the isosceles 3-body problem

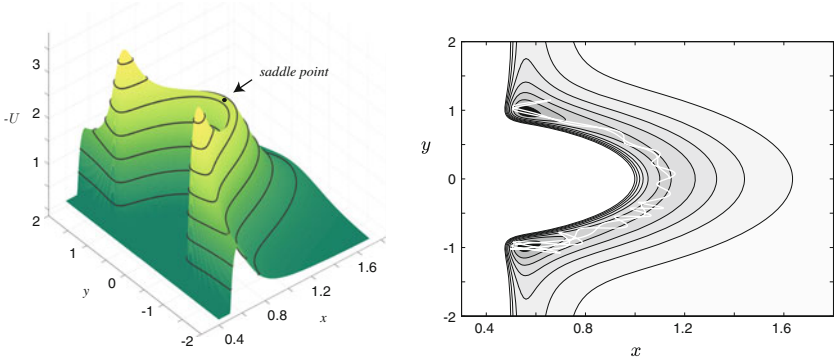


Fig. 1.22 *Left*: surface plot of the negative potential energy for the isosceles 3-body problem showing iso-potential curves. *Right*: the level sets of the potential energy function are shown for the isosceles 3-body problem, together with a snapshot of a constant energy trajectory (*light curve*)

kinetic energy is non-negative, we must have

$$2\hat{\varphi}_{\mathrm{LJ}}(\sqrt{x^2+y^2}) + \hat{\varphi}_{\mathrm{LJ}}(2x) \leq \mathrm{E}.$$

On the left side of Fig. 1.22 are shown the level curves of the potential energy function, with color gradations indicating the energy level between -3 and 0. (When $\mathrm{E} > 0$, the bodies eventually escape to infinity; $\mathrm{E} < -3$ is not attainable.) Superimposed on the level curves we have shown a short time trajectory (the wiggly white curve).

A surface plot of the **negative of** potential energy is shown on the left in Fig. 1.22. Arranging the three atoms in a collinear configuration ($y = 0$) the potential energy becomes $\hat{U} = \hat{U}(x) = 2\hat{\varphi}_{\mathrm{LJ}}(x) + \hat{\varphi}_{\mathrm{LJ}}(2x)$. Minimizing the potential in the collinear configuration allows us to determine the *saddle point* $(x_*, 0)$. Near this point, U decreases if we move in the $\pm y$ direction and increases if we move in the $\pm x$